CENTRE FOR DEVELOPMENT OF IMAGING TECHNOLOGY

TECHNOLOGY EXTENSION DIVISION

SYLLABUS

POST GRADUATE DIPLOMA IN COMPUTER APPLICATION (PGDCA Semester II)	
Scheme: 2023 Scheme Duration: 6 Months Eligibility: A Pass in Any Degree	
THEORY SUBJECTS	
PGDCA 201	Java Programming
PGDCA 202	Internet Technologies
PGDCA 203	Software Engineering & Project Management
PGDCA 204	Advanced Technologies
LAB SUBJECTS	
PGDCA 205	Lab I- Java Programming
PGDCA 206	Lab II- Web Technologies
PGDCA 207	Lab III- Mobile Programming
PGDCA 208	Project
ANNEXURE	
Assignments for Theory & Questions for Lab	

PGDCA 201 - JAVA PROGRAMMING
Theory Hours - 50 Hrs

Module I – Java Basics (10Hrs)

Learning outcome

- To understand the concept features of Java Programming
- To declare and identify variables, string handling functions and develop basic programming

Topics

Introduction to Java — — Features— Java Applications —Platform independent, Security — Bytecodes and JVM — Java versus C++ — Java Application —Java Applets-Java Languages — Data types in Java — Integral types—Byte,short,int,long — Floating points—float,double character — — Boolean typeVariables — Declaration of variable — Dynamic initialization — Scope of variables — Final variables — Type promotion—rules — Type casting-Literals — Integer,Floating point,Boolean,character and string literals Operators — Arithmetic Operators — Relational Operators — Increment and decrement Operators — Conditional Operators-Control Statements — IF, if………else, switch…..case — — Iteration-while, do while, for, break, continue-A Simple java program — Structure — The main( ) method — Command line arguments — Editing, Compiling and executing a java program Arrays — Declaring and creating arrays— —The new operator — — One-dimensional arrays — Multi dimensional arrays — Simple programs using arrays,read,print,sort,etc

Java Library — — String Handling –string class — — String Constructors — — String literals — Character extraction, methods, substring() — — Index methods, length() method — — String comparison method — — Conversion using value Of methods-String Buffer Class — StringBuffer construcors — — append(),insert(),and replace() methods

Module II – Object Oriented Programming (10Hrs)

Learning outcome

- To analyze the importance of Object Oriented Programming concept over structured programming and become familiar with the fundamental concepts in OOP like encapsulation, Inheritance and Polymorphism
- To learn concepts of inheritance to create new classes from existing one
- To familiarize the concepts of handling exceptions, packages and interfaces

Topics

Objects in Java — Class-syntax-class and method naming conversions — Class members, private, public and protected access — Creating objects — Object references — Defining Constructors — This keyword and its use — Program example using simple classes —— Final member and methods —— Static members and methods —— Overloading methods — Passing simple types and objects as method arguments Packages

OOPs Properties –Abstraction, Encapsulation, Inheritance, Polymorphism

Overview of packages in the java library — Java.lang— Java.io —Java.Util — Java.NET — Java.awt —— Java.sql — Defining a package — Classpath— Importing a package—import statement — Interfaces — Defining interface — Implementation interface — Interface variables — Extending interfaces Exception Handling —— Exception class hierarchy — Exception handling—try, catch, throw — Throws , finally ——Exception Handling

Inheritance —— Defining inheritance — Sub class, super class — Defining sub class- extends keyword —— Creating multi-level class hierarchy — Order of constructor execution — Super keyword — Method overriding —— Using final to prevent overriding-Polymorphism —— Abstract Classes-Multithreading —— Definition of thread —— Java:lang, —Thread class methods ——Creating a thread —— By extending thread class —— By implementing Runnable interface —Creating multiple threads —— Synchronizing threads — Synchronized methods —— Synchronized statement

Module III – File Handling and GUI Programming (10Hrs)

Learning outcome

- To understand the concepts of streams and file handling
- To learn the database connectivity and its programming
- To familiarize the concept of event handling used in GUI
- To understand the difference between AWT and SWING GUI programming concept

Topics

Concepts of Streams — Java.in package-io class hierarchy — —Input stream, Output stream, Data input Stream, Data Output Stream — — Reader and Writer classes — Using console input — — The file class — File input stream — File output stream — — File handling using File Input Stream and File Output Stream

Introduction to JDBC

Java awt package — Components and containers — — AWT class hierarchy-Layout manager—Flow layout, Grid layout, Border layout, Card layout —Event handling — Event model, Listeners — — Listener objects — Listener interfaces for most types of events-

SWING-Difference between awt and swing-Components hierarchy-Panes-Individual Swings components Jlabel, JButton, JTextField, JTextAres.

Introduction to Java Applets

Module IV - Introduction to J2EE (10Hrs)

Learning outcome

- To understand the concept of Java Web applications
- To learn the concept of JSP and servlet
- To develop basic programming using JSP and Servlet

Topics

Introduction to J2EE— Overview — Web applications — Enterprise applications —Sun J2EE Blueprints — Platform roles — Platform Contracts — Scalability and fault tolerance

Introduction to JSP-Life cycle of JSP-JSP API-JSP in Eclipse and other IDE's-Scripting Elements of JSP-Implicit Objects of JSP-Application of JSP

Introduction to Servlet-Servlet: What and Why?-Basics of Web-Servlet API-Servlet Interface-GenericServlet-HttpServlet-Servlet Life Cycle-Working with Apache Tomcat Server-Steps to create a servlet in Tomcat-How servlet works?-servlet in Myeclipse-servlet in Eclipse-servlet in Netbeans-Application of Servlet

Module V - Advanced Java (10Hrs)

Learning outcome

- To understand the concept of web applications using java,Spring,Spring MVC
- To build enterprise quality web applications using Spring MVC
- To learn the architecture of Hibernate and to know how to set up and configure Hibernate for a java project

Topics

Introduction to Spring-What is Core Container-Introduction to IOC-Types of DI-Setter DI vs Constructor DI-Resolving Constructor Confusion-Collection DI-Bean Inheritance-Collection Merging-Inner Beans-Using IDRef-Bean Aliasing-Bean Scopes-Inner Beans-Null String-Bean Auto wiring-Nested Bean Factories-Spring Core (3.0 Annotations)-Application of Spring

Introduction to MVC using Spring-What is MVC-Introduction to Maven-Application of MVC using Spring and Java

Introduction to Hibernate-Need for Hibernate-Hibernate and ORM (Object-Relation Mapping)-POJOs (Plain Old Java Objects) and the Data Layer-Hibernate Over Entity Beans-Understanding Hibernate Architecture

PGDCA 202 - INTERNET TECHNOLOGIES
Theory Hours - 50 Hrs

Module I - Computer Networking Technologies(10Hrs)

Learning outcome

- Knowledge about networking technologies and types of networks, devices used, security devices, topologies, nodes, mobile communication and latest technologies.

Topics

Introduction to Computer Networks - Evolution of Networking, Wired & Wireless Networking

Types of Networks - LAN (Local Area Network) , MAN (Metropolitan Area Network) , WAN (Wide Area Network) , PAN (Personal Area Network)

Network Devices - Hub, Switch, Router, Bridge, Gateway; Modem, Repeater, Access Point; Modem, Ethernet Card, RJ11, RJ45

Network Security Devices - Active Devices, Passive Devices, Preventative Devices, Unified Threat Management (UTM), Firewalls

Networking Topologies - Bus Topology, Star Topology, Ring Topology, Mesh Topology, Tree or Hybrid Topology

Nodes in a Networked Communication - Types of Nodes, MAC Address, IP Address, IPv4, IPv6

Connecting to Internet - Connecting the computer to the Internet

Types of connectivity – Dial-up; Wired broadband connectivity - ISDN, Cable Internet, DSL, Leased Line, FTTH; Wireless broadband connectivity - Mobile broadband, Wi-MAX, Satellite broadband

Internet Access sharing methods - LAN, Wi-Fi, Li-Fi

Services on Internet - World Wide Web (WWW), Browser, Web browsing, Search Engines, E-mail, Social media; Various stages of Data Transfer from Server to Client; Domain Name; Resolution; Mobile communication;
Generations in mobile communication - First Generation, Second Generation - GSM, GPRS, EDGE, CDMA, Third Generation – WCDMA, Fourth Generation, Fifth Generation;
Mobile communication services- SMS, MMS, GPS, Smart Cards
Mobile Operating System

Module II -Internet and e-commerce (10Hrs)

Learning outcome

- Explore different kinds of information available on the Internet, surfing safely and Searching information on the Internet
- Importance of online conference tools in daily life and its usage
- Understand e-learning and its importance
- Understand the concept of e-government and the associated benefits and drawbacks
- Gather knowledge on various National and State e-governance initiatives.
- Recognize the benefits and limitations of e-commerce
- Understand major e-commerce business models and their impact in the society

Topics

Concepts of Internet – Basics of internet connectivity -Applications of Internet –tools and troubleshooting– Intranet and extranet -Internet Services-communication services-Information retrieval services-Web services-World Wide Web-email– Web Browsing softwares – Search Engines – URL – Web Protocols- Domain name – IP Address – Surfing the web

Basics of electronic mail –concept, technology behind email, different email providers, Getting an email account – Sending and receiving emails – mail operations –email folders-applications of Emails-web mail-client mail-email security- email client(eg-outlook)

Online/Web/Mobile Conferences & Tools – introduction towards Audio and Video conferencing – Advantages & Disadvantages -Popular Video conferencing tools eLearning –What is eLearning, Features and uses, Different eLearning platforms in India, LMS

e-Commerce (Internet Commerce or electronic commerce)- What is e-Commerce - Features and Scope of e-Commerce –Traditional Commerce vs e-Commerce - e-Commerce Models (B2C, B2B, C2C, C2B, B2G - Benefits and limitations of e-Commerce – Major e-Commerce websites and its usage

What is E-Governance-objective, Scope, Benefits and outcomes of e-Governance - Pillars of e-Governance: People - Process - Technology - Resources, Types of interactions in e-Governance: Government to Citizen (G2C)- Government to Businesses (G2B)-Government to Government (G2G)-Government to Employees (G2E), Phases of e-Governance Maturity Model (Phase I: Information, Phase II: Interaction, Phase III- Transaction - Phase IV : Transformation),

e-Governance Infrastructures: Common Service Centre (CSC) - State Wide Area Network (SWAN) – National e-Governance Service Delivery Gateway (NSDG) - State Data Center (SDC)

Challenges to e-Governance, Security approaches to e-Governance

National level e-Governance initiative – Digital India initiative, Major services and projects implemented by union govt and state govt.

Module III - HTML, CSS and JavaScript (10Hrs)

Learning outcome

- To understand the working of a web application
- Understanding the use of HTML, CSS and JavaScript in websites
- To Develop websites

Topics

Introduction to web technology - Web application - Web server and Application server - Client and Server - Scripting languages

HTML and HTML5 - HTML Editors - Elements – Attributes - Headings - Paragraphs -Formatting - HTML Comments - Colors - Fonts - Hyperlinks - Images - Tables - Lists - Iframes - Marquee - Forms - Embedding Audio and video

XML introduction - XML Syntax and tags - XML Applications

CSS Introduction - Syntax - Selectors - External, Internal & Inline Style sheets - Backgrounds - Colors - Text - Fonts - Links - Border - Outline - Margin - Padding - Align - Positioning - Gradients

JavaScript introduction - Syntax - External & Internal usage - Variables - Operators - Functions - Events - Comparison - Condition - Loops - Dialog box - Form Validation.

Module IV- PHP Basics (10Hrs)
Learning outcome ➢ Understand working of PHP application ➢ To create web applications using PHP **Topics** ***Introduction to PHP*** - Advantages of PHP - Where to use PHP - Installing PHP - PHP Syntax - Variables - Echo and Print - Data Types –Strings - Constants - Operators - IF Statement - Switch statement - Loops – Functions – Arrays – PHP Forms – Date & Time – PHP File handling – PHP File Reading and Writing – Cookies – Sessions – Exceptions – PHP Classes and Objects – Constructor and Destructor – Access Modifiers – Inheritance – Abstract Classes – Static Methods.
Module V - PHP Advanced (10Hrs)
Learning outcome ➢ Understand working of PHP application ➢ To create web applications using PHP **Topics** ***Introduction to Database connectivity with PHP*** - Creating Database and Tables - Creating connection to the Database - Inserting Data into Database - Reading Data from Database- Updating Database Records - Deleting Database Records - Webpage database access. ***Content Management Systems (CMS)*** – Advantages of CMS– CMS Applications

PGDCA 203 - SOFTWARE ENGINEERING & PROJECT MANAGEMENT

Theory Hours - 50 Hrs

Module I - Introduction to Software Engineering & Process Models (10Hrs)

Learning outcome

- To plan and manage projects at each stage of the software development life cycle.
- Learn the concept of software engineering and its importance.

Topics

***Software Engineering Fundamentals*:** Introduction to Software Engineering: The evolving role of software; Software nature - changing nature of software; Principles of software engineering - Characteristics, Components, Applications; Software Process, Software Myths.

***Process Models*:** Generic Process Model; Prescriptive Process Models: The Waterfall, Incremental Process(RAD), V-Shaped Model, Agile Model**A**

Module II - Project Management, Estimations, Scheduling & SRS (10Hrs)

Learning outcome

- Understand Project Management principles while developing software.
- Gain extensive knowledge about the basic project management concepts, framework and the process models.
- To develop skills to manage the various phases involved in team management.
- To learn about the planning and risk management principles.

Topics

Software Project Management:- Software Project management ; Software Project - Types, Project Planning, Project scheduling Techniques ; Project management cycle ; Project Evaluation Review Technique (PERT), Gantt Charts, Network planning model; Resources Allocation

Software Project Estimation:- Problem-based estimation ; Process based estimation ; Project Planning ; Cost Estimation Model- COCOMO Model.

Project Scheduling:- Basic Concepts, Defining a Task Set for the Software Project, Defining Task Network

Requirements Engineering:- User and system requirements; A spiral view of the requirements engineering process; Software Requirements Specification (SRS) - The software requirements Specification document, The structure of SRS.

Project Risk Management : Software Risks, Risk Identification, Risk Projection, Risk Refinement, Risk Mitigation, Risks Monitoring and Management, Risk Control; Benefits of Risk management.

Software Development Team Structure :- Team Structures, Managing the Project Team; Change management: Dealing with Conflict & Resistance Leadership & Ethics;

Module III - Software Analysis and Design Engineering (10Hrs)

Learning outcome

- Understand software design concepts and their diagrammatic representation

Topics

Software Design: *Software Design* **-** Objectives, Process ; Coupling and Cohesion ; Functional Oriented Design - Data Flow Diagrams (DFD), Data Dictionary(DD), Entity-Relationship diagram(ERD).

Introduction to UML :- UML features and Advantages ; Object Oriented Analysis ; Principles ; UML diagrams - Use Case Models , Use Case Diagrams , Developing Use Case Diagrams for Typical Applications; Tools for creating UML diagrams.

Software Configuration Management: SCM- Introduction, need, basic configuration, management function.

Module IV- Software Testing & Quality Assurance (10Hrs)

Learning outcome

- Understand Software Quality Concepts.
- Familiarise to develop test plans to guide the testing stage of the software development lifecycle.

Topics

Software testing: - Introduction to Software Testing - Principles, Testing Life Cycle, types and levels of testing, test strategies; Program Correctness; Program verification and validation; Testing Automation & Testing Tools; Test metrics and Test reports.

Software quality assurance: - Software Reliability; Software Quality - Overview of ISO 9001, SEI-CMM,Six Sigma,IEEE

Software Change Management: Objectives; Version and Change Control; Auditing and Reporting

Module V- Software Maintenance, Reengineering & CASE Tools (10Hrs)

Learning outcome

- Understand the process associated with project closure to ensure that the project is closed in an orderly manner
- Define different kinds of CASE tools and the needs of the CASE tools

Topics

Maintenance & Reengineering:- Software Maintenance, Aims of Software Maintenance, Types of Software Maintenance, Software Maintenance Costs. Software Supportability, Reengineering, Software Reengineering, Reverse Engineering, Restructuring, Forward Engineering

Project Closeout:- Introduction; Project Closure - Reasons, Process, Performing a Financial Closure, Report

CASE Tools:- Introduction; Categories; Need; Characteristics;

PGDCA 204 - ADVANCED TECHNOLOGIES

Theory Hours - 50 Hrs

Module I - Enterprise Resource Planning (10Hrs)

Learning outcome

- To know the basics of ERP
- To understand the key implementation issues of ERP
- To know the Operation and Maintenance of an ERP

Topics

Enterprise Resource Planning(ERP)- Introduction-What is ERP-Overview and Evolution-Attributes-prerequisites-IT Requirement-Benefits of ERP – Conceptual Model of ERP--the Structure of ERP-Fundamental Technology-Functional modules of ERP software- Implementation of ERP: Life cycle; Implementation methodologies, transition Strategies-Success and failure in implementation – factors -Operation and Maintenance of an ERP system--Major ERP Packages

Tally as ERP software-importance-features and uses

Module II -Mobile Programming (10Hrs)

Learning outcome

- Gather basic knowledge in Mobile computing
- Develop Android Applications

Topics

Getting started with Android

Introduction to Android-Application Fundamentals-App Components-Features of Android-Architecture of Android-Timeline of android

Android Tools

How to install java development kit-Android studio-Adding sdk tools &packages to android studio-Configuring new project application-Configure with Real android device-Creating android virtual devices(AVDs)

Activities

Understanding Activities-Adding an Activity-Customizing Activity-Layouts in Activity-Activity Lifecycle-Android Manifest

User Interface

Linear Layouts-Relative Layouts-List view-Grid view--Frame layout--Input controls-Menus-Settings

Advanced User Interface development with material designing

Android design fundamentals-Surfaces-Toolbar-Navigation Drawer & Drawer Layout-Tab Layout-Recycler view as List View-Card view-View Pager

Intents

Android Intent-Start Activity via Intent-Send data to an Activity_android intent-Getting data from different activity-Android implicit intent

Fragments

Android Fragments-Fragment Lifecycle-Android ListFragment

Json introduction

Overview-Datatypes-Objects-Schema-Json with android

Advanced concepts

Notifications-Android Menu-Android Alarm Manager-Android Multimedia-Android Google Map-Sending Email-Sending Sms-Phone calls,common Android APIs, Sharing data between applications with Content Providers

Data Base in Android

SQ Lite introduction-Storing data using SQ Lite-Read/write from a data base

Webservice

Introduction to webservice-Webservice in android-Postman webservice testing-

Introduction to Networking Library

Introduction to Networking in android-Networking using library-Retrofit 2

, Publishing Android application in google playstore

Module III - Introduction to Artificial Intelligence & Machine Learning (10Hrs)

Learning outcome

- Understand the concepts of AI and ML
- Walkthrough various applications of AI in real world
- Understand the various ML Algorithms and its application in real life

Topics

Artificial Intelligence: Introduction; Meaning, Scope, and Stages of Artificial Intelligence; Stages in AI; Application of AI; Effects of Artificial Intelligence on Society

Machine Learning: Fundamentals of Machine Learning; Essential concepts of ML; Machine Learning – Process & Workflow; Idea of Machines learning from data, Classification of problem – Regression and Classification, Supervised and Unsupervised learning; Introduction to various ML Algorithms & Performance Metrics; Applications of Machine Learning

Module IV- Data Science and Data Analytics (10Hrs)

Learning outcome

- Become familiar with Data Science and Analytics concepts and the key steps in the process
- Understand different types of data structures, file formats, and sources of data
- Gather knowledge on various data analytic tools

Topics

Introduction to Data Science: - Evolution of Data Science; Data Science – Roles, Stages in a Data Science Project; Applications of Data Science in various fields; Data Security Issues; Data Analytics – Types, Phases, Quality and Quantity of data, Measurement, Exploratory data analysis; Business Intelligence; Introduction to Big Data.

Data Analytics: - Introduction, Data Summarization and Visualization; Linear and Nonlinear Regression; Model Selection; Classification, Logistic Regression; Clustering; Decision Trees

Data Analytics Tools: - Introduction to Data Analytic tools - R ; Tableau; RapidMiner; Power BI; other tools

Module V- Software Usability Engineering (10Hrs)

Learning outcome

- Understand the user interface characteristics of everyday objects affecting their usability.
- Understand the history of user interface design and its increasing important to software development.
- Understand development methodologies and lifecycle models for building user interface

Topics

Introduction -Definitions of human-computer interaction and user interface design; Importance of interface design in software design; Cost/benefit of good interface design

The Process - Usability Engineering: Stages in the development of user interfaces; Needs analysis; Systems analysis; User profiling; Preliminary design, including design notation; Rapid prototyping and interactive design; Formative evaluation techniques, including usability testing; Using standards and guidelines

The Product - Input and output devices; Dialog styles; Screen layout and design; User documentation; Evaluative testing

PGDCA Semester II

Detailed Syllabus

(Lab)

PGDCA 205 –LAB I – JAVA PROGRAMMING
Lab Hours - 120 Hrs

Module I

1. Basic Java Program to print a String Output
2. Program for Demonstrating the operation of Operators
3. Program for Demonstrating various data types of Java
4. Program for Demonstrating If Else Loop and Switch Loop
5. Programs for Demonstrating Single and Multiple Arrays
6. Programs for Demonstrating various String Functions
7. Write a program for reverse an array
8. Program for Demonstrating difference between String and String Buffer

Module II

1. Java Program for the Learning of Class, Member Elements, Member Functions, Object and implementing concept of OOPs
2. Java Program for implementing the concept of Inheritance
3. Java Program for demonstrating Run Time Polymorphism
4. Java Program for the Working of Abstract Class and Abstract Methods
5. Java program to demonstrate various types of exceptions
6. Java Program to create User defined exceptions
7. Java Program to demonstrate throws and finally
8. Java Program to demonstrate operator precedence

Module III

1. Program for Demonstrating various GUI Elements and Events (Button, textField, Label)
2. Program for Demonstrating various JDBC Operations with CRUD (Create, Read, Update, Delete)
3. Program for demonstrating prepared statement and JDBC
4. Program for demonstrating various file operations
5. Program for demonstrating Applet functionalities

Module IV

1. Program for demonstrating CRUD operations with JSP,Servlet
2. Program for demonstrating various properties of JSP,Servlet

Module V

1. Program for demonstrating Hibernate Operations and Functionalities
2. Program for demonstrating Spring Basic functionalities (IOC, Dependency Injections-CI/SI, Application Context)
3. Program for Demonstrating Spring MVC Functionalities

PGDCA 206 – LAB II- WEB TECHNOLOGIES
Lab Hours - 120 Hrs

1. Familiarization various videoconferencing tools
2. Familiarization email signature
3. Familiarization of online public services provided by state and central govt

4. Practicing basic HTML tags, font tags, paragraph styles, headings, lists
5. HTML programs that uses comments
6. HTML programming using table and its attributes
7. Tables in HTML, Frames in HTML, nested frames, Link and Anchor Tags
8. Creating marquee in HTML.
9. Programs that uses Iframe and forms
10. HTML programs to including graphics, video and sound in web pages
11. Programming exercise to create web page using HTML.
12. Practice basic XML programs
13. Practice basic CSS programs.
14. Practice CCS Programs based on CSS styling background-text, fonts, links
15. CSS Programs using border, outline, margin, padding
16. CSS programs using align, positioning and gradient.
17. Practice programs based on Javascript operators
18. Practice JavaScript programs based on conditional statements: if, if else, switch
19. Practice JavaScript programs based on loops
20. Practice JavaScript programs based on functions
21. Program to perform simple form validation using JavaScript
22. Perform programs that uses dialog boxes in JavaScript
23. PHP programs based on basic data types.
24. PHP programs based on various operators.
25. PHP programming exercise on Conditional Statements: if, if else, switch.
26. PHP programming using loops
27. PHP programming using functions

28. PHP programs using arrays
29. PHP programs based on file handling
30. PHP programs based on exceptions
31. PHP programs based on classes and objects
32. PHP programs based on constructor and destructor
33. PHP programs related with inheritance
34. PHP programs based on abstract classes
35. PHP programs based on static methods
36. PHP programming exercise to create a simple web page using scripting.
37. PHP programs connected with database and related operations

PGDCA 207 –LAB III- MOBILE PROGRAMMING
Lab Hours - 110 Hrs

1. Android program for calculator
2. Android program for sent data from one activity to second activity
3. Android program for open dialer in android
4. Android program for convert temperature from Celsius to Fahrenheit using radio button
5. Android program to check valid mobile number
6. Android program for display country name using List View
7. Android program for create contact menu
8. Android program for create audio player
9. Android program to find phone details
10. Android program for contact management system
11. Android program for Todo app creation using sqlite
12. Android program for simple text animation
13. Android program for alarm management system

PGDCA Semester II Assignments for Theory and Questions for Lab

PGDCA II Sem - Paper 1 (Theory)
PGDCA 201 - JAVA PROGRAMMING

Module I - Java Basics
Assignment

1. WAP to demonstrate the various escape sequences available in Java.
2. WAP to display the default values of all primitive data types of Java.
3. WAP to display a decimal, octal, and hexadecimal literal.
4. WAP to display the various data types available in Java and their respective size.
5. WAP to search a word inside a string.

Module II - Object Oriented Programming
Assignment

1. WAP to create a class Shape with a method calArea(double side) and three sub classes Cube, Circle, and Square that overrides the method calArea to calculate the respective area, using constructors and static variables. Also, include the final keyword to declare constants. (Hint: Pi=3.14)
2. WAP to overload an add function that add 2 values. Overload the add function on the primitive types.
3. WAP to create an abstract class Zoo with two methods body Color and food Type declared as abstract, and another method make Noise declared as non-abstract method. Also declare a static variable. Create classes that inherit the abstract class, with names of different animals. Within each class, assign the strength of the particular animal to the static variable.
4. WAP to implement a fixed-length version of an integer stack using interfaces. Declare the size of the stack within the interface.
5. WAP to demonstrate super, this, equals, getClass and toString and instance.Of operator.

Module III - File Handling and GUI Programming
Assignment

1. WAP to demonstrate the various controls available in AWT.
2. WAP to develop an applet that displays a simple banner.
3. WAP that creates a user interface to perform integer divisions. There should be 2 textfields to take in the operands, a button to perform the operation, and another textfield to show the result.
4. WAP to draw lines, rectangles and ovals.
5. WAP to demonstrate the various layouts.
6. WAP to connect to a database.
7. WAP to demonstrate scrollable result set with read only mode.
8. WAP to demonstrate database metadata.
9. WAP to get a JDBC Connection object using properties file.
10. WAP that creates a table Car in the database Vehicle with appropriate fields. If the table already exists, drop the previous table and create a new table. Insert some values in the fields and implement the CRUD operation

Module IV - Introduction to J2EE
Assignment

1. Write programs demonstrating the use of Scriptlet tag, Expression tag,Declaration Tag
2. Write a program for demonstrating a Login Form and Registration form in which Once successful login happens migration to Registration form takes place
3. Write a program for the File Upload using Servlet and cos jar
4. With example state the difference between Servlet Config and Servlet Context
5. Write a program for the URL rewriting using Servlet

Module V - Advanced Java
Assignment

1. Describe the structure of Hibernate program
2. Describe the structure of Various components of Spring MVC CRUD program
3. Explain the following Spring functionality
    1. IOC
    2. Dependency Injection
    3. Setter Injection
    4. Constructor Injection

PGDCA II Sem - Paper 2 (Theory)

PGDCA 202 - INTERNET TECHNOLOGIES

Module I - Computer Networking Technologies
Assignment

1. Explain the various broadband technologies available for Internet access
2. Write the disadvantages of social media. What are the different ways to avoid the disadvantages of social media?
3. Compare GSM and CDMA standards
4. Explain the advantages of a smart card.
5. Explain the advantages of 5G network

Module II - Internet and e-commerce
Assignment

1. Explain the idea of e-Office
2. Briefly explain the concept of cookies, browsing history, parental control
3. Explain the importance of google class room in modern education
4. What are the services provided by UMANG and Digi Locker?
5. What are the services provided by kerala.gov.in?
6. Write the importance of Citizen service portal
7. What are the benefits of e-Governance to society?

<u>Module III - HTML, CSS and JavaScript</u>
<u>Assignment</u>

1. Difference between HTML and HTML5
2. Explain any 5 HTML tags and their uses.
3. Explain the use of XML in application development
4. Write a short note on CSS and its uses in web application
5. Explain client server architecture with diagram

<u>Module IV - PHP Basics</u>
<u>Assignment</u>

1. Compare the advantages of PHP than other languages
2. Explain various operators in PHP with example
3. Explain various conditional and looping statements in PHP with example programs.
4. Explain how exceptions are handled in PHP
5. Briefly describe types of inheritance in PHP

<u>Module V - PHP Advanced</u>
<u>Assignment</u>

1. Create an application for registering user details into database.
2. Create an application to store telephone directory and perform search.
3. Create a contact us page to get user input data.
4. Write a short note on various Content Management Systems.
5. Compare WordPress, Joomla and Drupal

PGDCA II Sem - Paper 3 (Theory)

PGDCA 203 -SOFTWARE ENGINEERING & PROJECT MANAGEMENT

Module I - Introduction to Software Engineering & Process Models
Assignment

1. What are the principles of software engineering?
2. What is Agile Software Development? List various agile tools and techniques?
3. Difference between Agile and traditional Methodology
4. When or on what circumstance do the following SDLC models need to be used:
    a. Iterative Model
    b. V-Shaped Model
    c. WaterFall model
    d. Agile Model
    e. Spiral Model
    f. BigBang Model

Module II - Project Management, Estimations, Scheduling & SRS
Assignment

1. What is project management? Briefly describe the project management framework, providing examples of stakeholders, knowledge areas, tools and techniques, and project success factors.
2. What is Scrum in Project Management? Explain in detail
3. Describe any two approaches to develop a strategy for change.
4. Explain the purpose of a Software configuration management plan.
5. Prepare a SRS document for an e-commerce website.

Module III - Software Analysis and Design Engineering
Assignment

1. List 9 modeling diagrams that are frequently used and explain in detail with examples
2. List various UML Tools
3. Explain different types and Components of Data Flow Diagram (DFD)
4. Difference between Data Dictionary and ER model using an example

Module IV - Software Testing & Quality Assurance
Assignment

1. What are the main differences between Software Quality Assurance, Quality Control, and Software Testing?
2. Explain unit testing and Code Inspection
3. Describe some of the steps for administrative closure.
4. If you have a test plan with hundreds of test cases how would you decide what tests need to be automated and what to cover in manual testing?
5. How do version control systems ensure that two software developers do not attempt the same change at the same time?
6. What is the difference between
    1. a Test Plan and a Use Case?
    2. a Test Plan and Test Strategy?
    3. CMMI and ISO?

Module V - Software Maintenance, Reengineering & CASE Tools
Assignment

1. What are the principal factors that affect the costs of system re-engineering?
2. What is the difference between reverse engineering and re-engineering?
3. How do CASE tools support different phases?
4. Define metric. Discuss seven core metrics for project control and process instrumentation with suitable examples.
5. Suppose a project was estimated to be 600 KLOC. Calculate the effort and development time for each of the three models i.e., organic, semi-detached & embedded.
6. A project size of 350 KLOC is to be developed. Software development team has average experience on similar types of projects. The project schedule is not very tight. Calculate the Effort, development time, average staff size, and productivity of the project.

PGDCA II Sem - Paper 4 (Theory)
PGDCA 204 - ADVANCED TECHNOLOGIES

Module I - Enterprise Resource Planning Assignment

1. ERP system allow simultaneous access to the same data for planning and control. Discuss in detail.
2. What are the reasons for the explosive growth of the ERP market?
3. What are the direct benefits of ERP systems?
4. Explain in detail the factors that are critical for the success of the ERP implementation?
5. Define Enterprise and ERP.
6. What are ERP packages? Explain major ERP packages in detail

Module II - Mobile Programming Assignment

1. What is an Android and Android Architecture? Explain the features of Android Architecture.
2. Describe Activity Lifecycle with diagram.
3. What is the fundamental UI design in Android?
4. What does an intent filter do? Mention different intent types.
5. Differentiate between :
   a) Activity and Widget
   b) Linear, Relative and Absolute Layouts
   c) Frame Layout and Table Layout
6. What is SOLite? Elaborate on the important features of SQLite.
7. Explain Sensors in Android.
8. Explain the AndroidManifest.xml file and why do you need this?
9. Discuss various steps involved in publishing an Android App on Google PlayStore.

Module III - Introduction to Artificial Intelligence & Machine Learning
Assignment

1. What is Artificial Intelligence? Give an example of where AI is used on a daily basis.
2. Explain various domains of Artificial Intelligence in detail.
3. What is Machine Learning? Discuss various types of Machine Learning.
4. How is Machine learning related to Artificial Intelligence?
5. Will artificial intelligence in near future reach the state of self-perfecting intelligence. Discuss

Module IV - Data Science and Data Analytics
Assignment

1. Explain about roles and stages in data science project.
2. Explain about Data Cleaning and Sampling.
3. What is the difference between data analytics and data science?
4. What is logistic regression? State an example where you have recently used logistic regression.
5. How do you approach solving any data analytics based project?
6. Discuss about Big Data-Characteristics and applications? Why is Big Data Important?
7. Explain in R Language the following

   a) Different concepts of arrays b)Lists c)Matrices
8. What are the different datatypes in Tableau?

Module V - Software Usability Engineering
Assignment

1. What is human computer interface?
2. What is Usability Testing? What is the purpose of usability testing?
3. State the impact of human characteristics on user interface design?
4. Discuss the benefits of a well-designed user interface design.
5. Explain the importance of user documentation

PGDCA II Sem - Paper 5 (LAB)
PGDCA 205 – JAVA PROGRAMMING

1. Create a Hello World Application in Java
2. Write a Program for Demonstrating Addition, Subtraction, Multiplication and Division operations in Java for integer, float, double and short variables
3. Write a program for initializing various data types of Java
4. Write a program for ATM functionality using If Else loop and Switch loop
5. Write a program to sort one dimensional array
6. Write a Program for demonstrating various string functionalities
7. Write a program for reverse an array
8. Write a program for Matrix multiplication
9. Write a Program for checking whether a given String is Palindrome or not
10. Write a program for finding difference between String and String **Buffer**
11. Create a java program using class and object in java for the addition of 2 numbers
12. Create 2 java classes and demonstrate the concept of Inheritance
13. Create a class and Object in java for demonstrating Run Time Polymorphism
14. Create a class named 'Member' having the following members:

    Data members

    1 – Name, 2 – Age , 3 - Phone number ,4 – Address , 5 – Salary

    It also has a method named 'printSalary' which prints the salary of the members. Two classes 'Employee' and 'Manager' inherits the 'Member' class. The 'Employee' and 'Manager' classes have data members 'specialization' and 'department' respectively. Now, assign name, age, phone number, address and salary to an employee and a manager by making an object of both of these classes and print the same.

15. Create an abstract class 'Parent' with a method 'message'. It has two subclasses each having a method with the same name 'message' that prints "This is first subclass" and "This is second subclass" respectively. Call the methods 'message' by creating an object for each subclass.
16. Write a program to demonstrate various types of exceptions
17. WAP to divide a number by zero.
18. Write a Program to create User defined exceptions
19. WAP to demonstrate throws and finally
20. WAP to convert a given Fahrenheit value into Celsius value to demonstrate operator precedence
21. WAP that simulates a traffic light.
22. WAP for handling mouse events.
23. WAP to simulate a calculator using Swing package
24. WAP to read text from a file and display the contents.
25. WAP to append data to an existing file.
26. WAP to execute and read select queries using JDBC
27. WAP to demonstrate statement, prepared statement, callable statement, and batch update
28. WAP to get primary key value (auto-generated keys) from inserted queries using JDBC.
29. WAP to create a moving Object using Applet
30. WAP for the creating new directory, file, delete file in java

31. See the following image and

a) Create a CRUD application for the above in JSP

b) Create a CRUD application for the above in Servlet

32. WAP to demonstrate Implicit objects in JSP
33. WAP for the Request Dispatcher in Servlet
34. WAP for the http session in Servlet
35. WAP for the CRUD applications using Hibernate in Core Java
36. WAP for demonstrating IOC and Dependency Injection in Spring
37. WAP for the CI and SI properties of Spring
38. WAP for the CRUD Application using Spring MVC and hibernate
39. WAP for demonstrating a CRUD application using Spring MVC and Maven

PGDCA II Sem - Paper 6 (LAB)
PGDCA 206 – WEB TECHNOLOGIES

1. Print two lists with different versions of Windows OS and Android. One list should be an ordered list, the other list should be an unordered list
2. Create a web page with link name "C-DIT" and reference goes to www.cdit.org
3. Display five different images. Skip two lines between each image. Each image should have a title.
4. Create a Contact Us Form for a website using HTML
5. Create a Registration Page for registering student details using HTML
6. Write a JavaScript program to find the characters with in a string.
7. Write a JavaScript program to print numbers from 1 to 100 using for loop.
8. Write a JavaScript program to print first 100 counting numbers using while loop.
9. Create a login page and perform form validation using JavaScript
10. Create an XML Syntax to store Book information
11. Create a Google Search page using HTML and CSS
12. Write a PHP program to find the given number is even or odd
13. Write a PHP program to find the number is prime or not
14. Write a PHP program to print the table of a number
15. Write a PHP program to print factorial of a number.
16. Write a PHP program to check Armstrong number.
17. Write a PHP program to check palindrome number
18. Write a PHP program to print Fibonacci series without using recursion and using recursion.
19. Write a PHP program to reverse given number
20. Write a PHP program to find if the given year is leap year or not.
21. Write a PHP program to read a file and print it
22. Create a website registration page to store student details into the database using PHP

23. Print all the student details from database table
24. Create a login page and perform login action
25. Design and develop dynamic web application using PHP for employee data with insert, delete, view and update operations
26. Design and develop Library Management system for storing book details using PHP

PGDCA II Sem - Paper 7 (LAB)
PGDCA 207 – MOBILE PROGRAMMING

1. Android program for calculator
2. Android program for sent data from one activity to second activity
3. Android program for open dialer in android
4. Android program for convert temperature from Celsius to Fahrenheit using radio button
5. Android program to check valid mobile number
6. Android program for display country name using List View
7. Android program for create contact menu
8. Android program for create audio player
9. Android program to find phone details
10. Android program for contact management system
11. Android program for Todo app creation using sqlite
12. Android program for simple text animation
13. Android program for alarm management system